

The Hadamard-diagonal row-column module is C^* -like exactly at $p = 2$

Candidate full solution for arXiv:2505.19471, Example 5.2

11 August 2026

Status. Candidate full solution; likely valid, pending human review. The proof is finite-dimensional and analytic. The included code only checks its explicit identities and is not used as proof.

1 Source question

Calin, Cartwright, Coffman, Delfín, Girard, Goldrick, Nerella and Wu, *C^* -like modules and matrix p -operator norms*, arXiv:2505.19471, Example 5.2 and the paragraph following it (source PDF, p. 18), ask whether their simultaneously Hadamard-diagonalizable unital algebra gives a non- C^* -like row-column module for $p \in (1, \infty) \setminus \{2\}$ [1].

(4) $(1, 0, 0, 1)$: Again, rotate so that $v_1 = 0$. The problem reduces to

$$\max_{\theta} |3 - e^{i\theta}| + |1 + e^{i\theta}| = \max_{\theta} \sqrt{10 - 6 \cos \theta} + \sqrt{10 + 6 \cos \theta} = 2\sqrt{10},$$

maximized at $\theta = \frac{\pi}{2}$.

This proves the claim. ■

Hence, using the claim we get at once

$$\max\{\|(\mathbf{b} \mid \mathbf{a})\|_A : \mathbf{b} \in \text{Ba}(M_{1,2}^1(A))\} = \sqrt{10} < 4 = \|\mathbf{a}\|_{M_{4,2}^1(\mathbb{C})}.$$

Therefore, C^* -likeness condition (4) does not hold, showing that $(M_{1,2}^1(A), M_{2,1}^1(A))$ is not a C^* -like module.

We do not know yet whether Example 5.2 works for $p \in (1, \infty) \setminus \{2\}$. Of course, for $p = 2$, A is a C^* -algebra and therefore the example does not work (see Lemma 2.13). We are currently working with both numerical computations and interpolation techniques to investigate Example 5.2 when $p \in (1, \infty) \setminus \{2\}$.

Funding. All authors were partially supported by the Department of Mathematics at the University of Colorado, Boulder via the Summer 2024 REU(G) program.

Figure 1: The exact open-question paragraph and its immediate conclusion in the source paper, p. 18.

The answer is yes for every such p . More precisely, the source column is a witness below 2, and its adjoint row is a witness above 2.

2 Definitions and statement

For $1 < p < \infty$, let

$$H = 2^{-1/2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad A_p = \{H \operatorname{diag}(\lambda_1, \lambda_2) H : \lambda_1, \lambda_2 \in \mathbb{C}\} \subseteq \mathcal{L}(\ell_p^2),$$

with its inherited p -operator norm. Put

$$a_1 = \frac{1}{2} \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}, \quad a_2 = \frac{1}{2} \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix}, \quad T = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}.$$

Thus $T \in X_p := M_{2,1}^p(A_p)$, while $T^* = [a_1 \ a_2] \in Y_p := M_{1,2}^p(A_p)$. The pairing is matrix multiplication, $(B \mid C)_{A_p} = BC$.

Theorem 1 (Full classification of Example 5.2). *For $1 < p < \infty$, the row-column module (Y_p, X_p) over A_p is C^* -like if and only if $p = 2$. More explicitly:*

1. if $1 < p < 2$, then

$$\max_{B \in \operatorname{Ba}(Y_p)} \|BT\|_{A_p} < \|T\|_{X_p};$$

2. if $2 < p < \infty$, then

$$\max_{C \in \operatorname{Ba}(X_p)} \|T^*C\|_{A_p} < \|T^*\|_{Y_p}.$$

Consequently the first norm-recovery axiom fails below 2, and the second one fails above 2.

3 Proof intuition

Below 2, the source column has only two norming input directions: the coordinate axes. If a structured row contraction recovered its norm, dual smoothness would force the adjoint column contraction to carry a unit vector to the unique support functional of Te_1 . Simultaneous Hadamard diagonalization makes that adjoint column unique once the unit vector is chosen. Its norm-attainment equations leave only four projective choices: $e_1, e_2, (1, i), (1, -i)$. Strict Clarkson inequalities exclude the coordinate directions, and a positive second variation excludes the two complex phase directions. Compactness then makes the recovery gap strict.

Above 2, no new optimization is needed. Taking Banach adjoints identifies columns at p with rows at the conjugate exponent p' , preserves all operator norms, and reverses products. Thus the strict failure below 2 becomes strict failure of the other C^* -likeness axiom above 2.

4 Scalar lemmas

For a unit vector $z \in \ell_r^m$, write $j_r(z) = (|z_k|^{r-2} z_k)_k \in \ell_{r'}^m$. We use the Hermitian dual pairing, linear in the first variable.

Lemma 1 (Scalar Clarkson inequalities). *For $z, w \in \mathbb{C}$,*

$$|z + w|^r + |z - w|^r \begin{cases} \leq 2(|z|^r + |w|^r), & 1 < r < 2, \\ \geq 2(|z|^r + |w|^r), & 2 < r < \infty. \end{cases}$$

In either non-Hilbertian range the inequality is strict when $zw \neq 0$.

Proof. Put $a = |z|$, $b = |w|$, and write $|z \pm w|^2 = a^2 + b^2 \pm 2ab \cos \theta$. If $1 < r < 2$, concavity of $t^{r/2}$ shows that the sum of the two $r/2$ -powers is at most $2(a^2 + b^2)^{r/2}$, which is at most $2(a^r + b^r)$. Both inequalities are strict when $ab > 0$. For $r > 2$, convexity reverses both comparisons. $\square$

Lemma 2 (The exceptional-phase inequality). *Let $s > 2$ and $q = 3^{1/(s-1)}$. Then*

$$(s-1)(q^{s-2} + q^2) > q^s + 1.$$

Proof. Set $L = \log 3$, so $1 < q < 3$, $s-1 = L/\log q$, and $q^{s-1} = 3$. After multiplying by $\log q$, the desired inequality is

$$h(q) := L(3/q + q^2) - (3q + 1) \log q > 0.$$

We have $h(3) = 0$. Moreover $-h'(q) = R(q)$, where

$$R(q) = 3 \log q + 3 + q^{-1} - L(2q - 3q^{-2}).$$

For $1 \leq q \leq 3$,

$$R'(q) = 3q^{-1} - q^{-2} - 2L - 6Lq^{-3} < 2 - 2L < 0.$$

Here $3q^{-1} - q^{-2} \leq 2$ and $L > 1$. Hence

$$R(q) \geq R(3) = \frac{10 - 8 \log 3}{3} > 0;$$

the last inequality follows, for example, from $\log 3 < 5/4$. Thus $h' < 0$ on $(1, 3)$, and $h(q) > h(3) = 0$. $\square$

5 Failure for $1 < p < 2$

Fix $1 < p < 2$ and put $s = p' = p/(p-1) > 2$. For $x, y \in \mathbb{C}$, Lemma 1 gives

$$\begin{aligned} \left\| T \begin{pmatrix} x \\ y \end{pmatrix} \right\|_p^p &= 2^{-p} (|3x + y|^p + |3x - y|^p + |x + 3y|^p + |-x + 3y|^p) \\ &\leq 2^{1-p} (3^p + 1) (|x|^p + |y|^p). \end{aligned}$$

Equality holds at the coordinate axes, and is strict if $xy \neq 0$. Therefore

$$\|T\| = N := 2^{1/p-1} (3^p + 1)^{1/p}, \tag{1}$$

and the only projective norming inputs are e_1, e_2 .

Suppose, toward a contradiction, that $\max_{B \in \text{Ba}(Y_p)} \|BT\| = N$. Finite-dimensional compactness provides a contraction $B \in Y_p$ and a unit vector at which equality holds. The preceding paragraph makes that vector a coordinate axis. The two cases are equivalent. Indeed, if $S = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \in A_p$ and $P = \text{diag}(S, S)$, then $PTe_1 = Te_2$, while P is an isometry and $BP \in Y_p$ whenever $B \in Y_p$. We may therefore use e_1 .

Let

$$\eta = N^{-1}Te_1 = \frac{1}{2N}(3, 1, 3, -1)^T, \quad q = 3^{p-1},$$

and choose the unit norming functional $\phi = (u, v)^T \in \ell_s^2$ of $B\eta$. With $C = B^* : \ell_s^2 \rightarrow \ell_s^4$, equality in the contraction chain yields

$$C\phi = g := j_p(\eta) = D^{-1}(q, 1, q, -1)^T, \quad C^*\eta = j_s(\phi), \tag{2}$$

where

$$D^s = 2(q^s + 1). \quad (3)$$

Indeed, the first equality is the unique-support-functional equality, while the second follows by applying the same equality to the contraction C , which maps the unit vector ϕ to the unit vector g .

Write the two blocks of C as

$$C_1 = H \operatorname{diag}(\mu_1, \mu_2) H, \quad C_2 = H \operatorname{diag}(\mu_3, \mu_4) H,$$

and put $\alpha = (u + v)/\sqrt{2}$, $\beta = (u - v)/\sqrt{2}$. The first equality in (2) gives

$$\mu_1 \alpha = \frac{q+1}{\sqrt{2D}}, \quad \mu_2 \beta = \frac{q-1}{\sqrt{2D}}, \quad \mu_3 \alpha = \frac{q-1}{\sqrt{2D}}, \quad \mu_4 \beta = \frac{q+1}{\sqrt{2D}}. \quad (4)$$

In particular, $\alpha\beta \neq 0$, and C is uniquely determined by ϕ . Insert (4) into the second equality in (2). After cancelling one common positive constant, the two Hadamard coordinates say

$$\frac{1}{\bar{u} + \bar{v}} = \kappa(|u|^{s-2}u + |v|^{s-2}v), \quad \frac{1}{\bar{u} - \bar{v}} = \kappa(|u|^{s-2}u - |v|^{s-2}v)$$

for a nonzero common scalar κ . Adding and subtracting these equations shows that, when $uv \neq 0$,

$$-\frac{\bar{v}}{\bar{u}} = \frac{|v|^{s-2}v}{|u|^{s-2}u}. \quad (5)$$

Taking moduli in (5) gives $|u| = |v|$; taking phases then gives $v/u = \pm i$. If one coordinate vanishes, ϕ is a coordinate vector. Up to a unimodular scalar, only

$$e_1, \quad e_2, \quad 2^{-1/s}(1, i)^T, \quad 2^{-1/s}(1, -i)^T \quad (6)$$

remain.

We now show that the unique C in each case has norm strictly greater than one, contradicting $\|C\| = \|B\| \leq 1$.

If $\phi = e_1$, equations (2) and the block symmetry give

$$C = D^{-1} \begin{bmatrix} q & 1 \\ 1 & q \\ q & -1 \\ -1 & q \end{bmatrix}.$$

Applying this matrix to $(1, 1)^T$, using (3), and invoking the strict $s > 2$ case of Lemma 1, gives

$$\frac{\|C(1, 1)^T\|_s^s}{\|(1, 1)^T\|_s^s} = \frac{(q+1)^s + (q-1)^s}{2(q^s + 1)} > 1.$$

The case e_2 is the same after swapping coordinates.

It remains to exclude $2^{-1/s}(1, i)^T$; complex conjugation handles the minus sign. Let $\phi_+ = (1, i)^T$, $\phi_- = (1, -i)^T$, and let C_0 be the unique structured column satisfying $C_0\phi_+ = g$. Directly from (4),

$$C_0 = \frac{1}{2D} \begin{bmatrix} q-i & 1-iq \\ 1-iq & q-i \\ q+i & -1-iq \\ -1-iq & q+i \end{bmatrix}, \quad C = 2^{1/s}C_0.$$

For real t near zero set $\xi_t = \phi_+ + it\phi_-$. A direct calculation gives

$$\begin{aligned} F(t) &:= \|C\xi_t\|_s^s - \|\xi_t\|_s^s \\ &= \frac{(q+t)^s + (q-t)^s + (1+qt)^s + (1-qt)^s}{q^s + 1} - 2(1+t^2)^{s/2}. \end{aligned}$$

Thus $F(0) = F'(0) = 0$, while

$$F''(0) = 2s \left(\frac{(s-1)(q^{s-2} + q^2)}{q^s + 1} - 1 \right) > 0$$

by Lemma 2. Hence $F(t) > 0$ for all sufficiently small nonzero t , so again $\|C\| > 1$. This exhausts (6) and proves the first assertion of Theorem 1.

6 Adjoint duality and the range $2 < p < \infty$

Let now $2 < p < \infty$, and put $r = p' \in (1, 2)$. Banach adjoints give isometric bijections

$$X_r \longleftrightarrow Y_p, \quad Y_r \longleftrightarrow X_p,$$

because every matrix in the Hadamard-diagonal algebra has its adjoint in the same algebra, and $\|S\|_{\ell_r \rightarrow \ell_r} = \|S^*\|_{\ell_p \rightarrow \ell_p}$. They also reverse products. Therefore the already proved strict inequality at r implies

$$\begin{aligned} \max_{C \in \text{Ba}(X_p)} \|T^*C\|_{A_p} &= \max_{B \in \text{Ba}(Y_r)} \|(BT)^*\|_{A_p} \\ &= \max_{B \in \text{Ba}(Y_r)} \|BT\|_{A_r} < \|T\|_{X_r} = \|T^*\|_{Y_p}. \end{aligned}$$

This proves the second assertion. At $p = 2$, H is unitary and A_2 is a C^* -algebra, so the row-column module is C^* -like by the standard C^* -module norm recovery (also Lemma 2.13 of the source paper [1]). Theorem 1 follows.

7 Verification and novelty status

The analytic audit is recorded in `verification_report.md`. The script `code/check_identities.py` checks the normalization identities, the two strict-expansion mechanisms, the scalar monotonicity margin, and random samples of the source-column norm for representative exponents. It is a sanity check, not part of the proof.

A bounded novelty check on 11 August 2026 searched the run's registry, solution, attempt and proof-gap indexes for arXiv:2505.19471, the exact title, “ C^* -like module”, “Example 5.2”, and the Hadamard/simultaneously diagonalizable formulation. Web searches for the exact title and exact open question found the arXiv paper and its 2026 Annals of Functional Analysis publication, whose p. 18 still states the question as open, but no separate answer. This is not a systematic citation-database review; novelty confidence is therefore moderate, while the source-question match is high.

8 Limitations and reviewer focus

The theorem concerns exactly the two-summand row-column module and algebra of Example 5.2. It does not classify arbitrary simultaneously diagonalizable subalgebras or higher-dimensional

Hadamard changes of basis. The most useful human checks are: (i) the equality-case passage from (2) to (5); (ii) the normalization $D^s = 2(q^s + 1)$; and (iii) the adjoint identification of the two module norm-recovery axioms.

References

- [1] A. Calin, I. Cartwright, L. Coffman, A. Delfín, C. Girard, J. Goldrick, A. Nerella and W. Wu, *C^* -like modules and matrix p -operator norms*, Ann. Funct. Anal. (2026), DOI: 10.1007/s43034-025-00492-8; arXiv:2505.19471.